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PID Motor Controller

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EN 2111 Electronic Circuit Design

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Abstract This project presents the design and implementation of an analog PID-based motor speed controller capable of maintaining a constant DC motor speed under varying load conditions. The system derives a feedback voltage proportional to motor speed through a frequency-to-voltage converter, compares it against a reference speed input, and processes the resulting error using an analog PID network built entirely from operational amplifiers. The controller output is converted into a duty-cycle-modulated PWM signal via a triangular-wave generator and comparator stage, which then drives the motor through an TIP31C Transistor. The system was simulated and tested to verify accurate, stable, and fast speed regulation across the target operating range.

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1 Introduction and Functionality

Our project, “**PID Motor Controller**”, aims to demonstrate closed-loop analog control of DC motor speed using a Proportional-Integral-Derivative (PID) compensator built from op-amp circuits. The controller continuously compares the actual motor speed, obtained via feedback, against a user-defined reference speed, and adjusts the drive signal to the motor so that the speed error is driven toward zero, even in the presence of load disturbances.

1.1 Requirements

- **Core Function:** Maintain constant motor speed despite load variations
- **Reference Input:** Adjustable reference speed voltage
- **Feedback:** Motor speed sensed via encoder and converted to a proportional voltage
- **Control:** Analog PID compensation (Proportional, Integral, Derivative)
- **Actuation:** PWM-based motor drive through a Transistor
- **Power Supply:** Stable dual supply ($\pm 12\text{V}$) and 6 V motor supply

1.2 Design Considerations

- **Analog Principles:** Realize P, I, and D actions using op-amp integrator, differentiator, and summing/differential amplifier stages
- **Feedback Path:** Convert motor speed (frequency) into a clean, ripple-free voltage suitable for comparison
- **Error Handling:** Accurately compute the difference between reference and feedback signals
- **PWM Generation:** Generate a stable triangular carrier wave and compare it against the PID output to produce a duty-cycle-modulated drive signal
- **Driver Stage:** Provide sufficient current and isolation between the control circuit and the motor
- **Tuning:** Allow adjustment of K_p , K_i , and K_d via potentiometers to shape transient response
- **Stability:** Minimize overshoot and oscillation while keeping steady-state error low

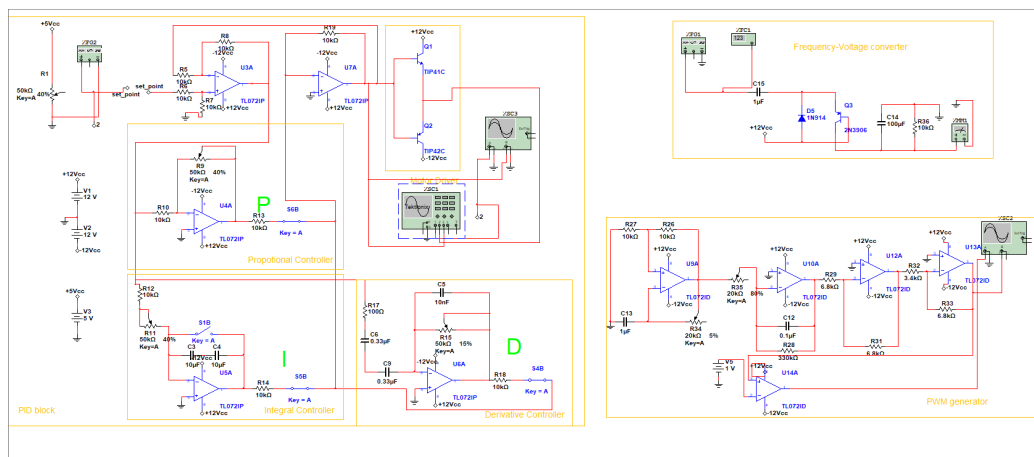


Figure 1: Overall System Block Diagram

2 System Architecture

The overall system consists of three main stages: Feedback Generation, the PID Controller, and the Driver Stage, as illustrated in the block diagram below.

2.1 Feedback Generation (F-V Converter)

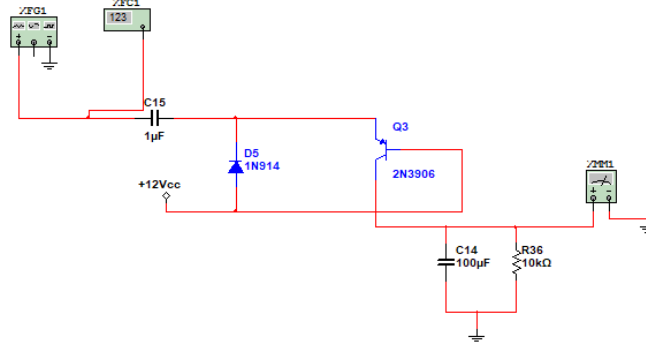


Figure 2: Frequency-to-Voltage Converter Circuit

Principle of Operation

- The motor's built-in magnetic encoder produces a pulse train whose frequency is proportional to motor speed.
- This pulse train is AC-coupled through capacitor C_{15} and shaped by diode D_5 and transistor Q_3 to produce fixed-width current pulses for every encoder edge.
- The pulses are averaged by the low-pass filter formed by C_{14} and R_{36} , producing a DC voltage proportional to pulse repetition rate.
- Since pulse frequency is directly proportional to motor speed, the resulting averaged voltage is also proportional to motor speed:

$$V_{fb} \propto f_{enc} \propto \omega_{motor}$$

- This feedback voltage V_{fb} is fed back to the PID controller for comparison against the reference voltage.

Derivation For a stable output level, the charge transferred from C_{15} to C_{14} must equal the charge lost each cycle from C_{14} through R_{36} .

$$(V_{IN} - V_{D5} - V_{BE,Q3}) C_{15} = \frac{T \cdot V_{OUT}}{R_{36}}$$

$$V_{OUT} = \frac{C_{15} R_{36}}{T} (V_{IN} - V_{D5} - V_{BE,Q3})$$

$$V_{OUT} = f \cdot C_{15} \cdot R_{36} \cdot (V_{IN} - V_{D5} - V_{BE,Q3}) \quad \text{where } f = \frac{1}{T}$$

2.2 PID Controller

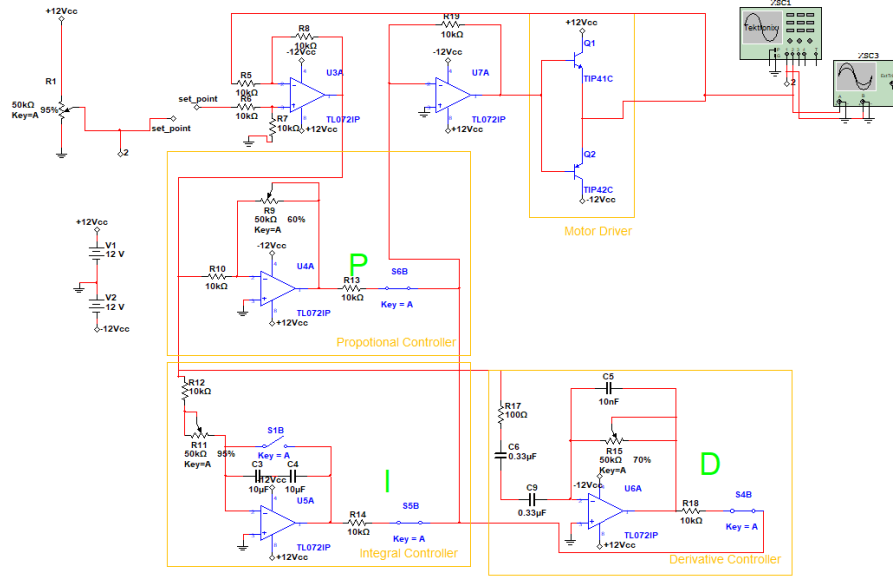


Figure 3: Analog PID Controller Circuit

Principle of Operation The controller computes the control signal $u(t)$ from the instantaneous error $e(t) = V_{\text{ref}} - V_{\text{fb}}$ according to the standard PID law:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

2.2.1 Error Detection

A differential amplifier stage subtracts the feedback voltage from the reference voltage to generate the error signal, following the same principle as a standard op-amp subtractor:

$$e(t) = V_{\text{ref}} - V_{\text{fb}}$$

2.2.2 Proportional Path

The error signal is scaled by an adjustable-gain inverting amplifier stage, with the gain potentiometer setting K_p :

$$V_P = -\frac{R_9}{R_{10}} e(t) = K_p e(t)$$

2.2.3 Integral Path

- Apply KCL at the inverting node of the integrator op-amp (virtual ground):

$$\frac{e(t)}{R_{11,12}} + C_{3,4} \frac{dV_{\text{out}}}{dt} = 0$$

- Solving for V_{out} :

$$V_I(t) = -\frac{1}{R_{Eq} C_{Eq}} \int_0^t e(\tau) d\tau = K_i \int_0^t e(\tau) d\tau$$

2.2.4 Derivative Path

- Apply KCL at the inverting node of the differentiator op-amp, where the input current through capacitor C_6 and C_9 equals the feedback current through R_{15} :

$$C_{6,9} \frac{de(t)}{dt} = -\frac{V_{out}}{R_{15}}$$

- Solving for V_{out} :

$$V_D(t) = -R_{15}C_{Eq} \frac{de(t)}{dt} = K_d \frac{de(t)}{dt}$$

2.2.5 Summing Stage

The three terms V_P , V_I , and V_D are combined using an inverting summing amplifier to produce the final control output:

$$u(t) = V_P + V_I + V_D$$

2.3 PWM Generation

Driving the motor with a duty-cycle-modulated square wave rather than a continuously variable analog voltage, allows the use of an efficient switching motor driver stage, minimizing power dissipation in the driver while still delivering an effectively analog-controlled average voltage to the motor.

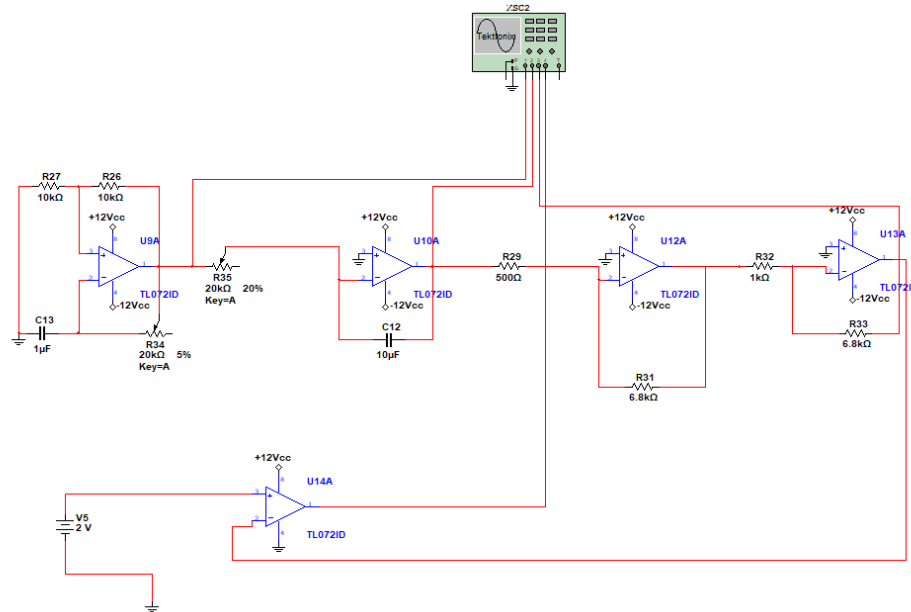


Figure 4: PWM Generation Circuit

Principle of Operation PWM generation is achieved using a triangular wave generator (formed from a relaxation oscillator and an integrator stage) followed by a voltage comparator.

1. **Triangular Wave Generator:** A square-wave relaxation oscillator drives an op-amp integrator, producing a linear triangular carrier waveform at a fixed switching frequency, set by R_{R26} , R_{34} , and C_{13} .

2. **Comparator:** The triangular carrier is compared against the PID control voltage $u(t)$. Whenever $u(t)$ exceeds the instantaneous carrier level, the comparator output switches high, producing a rectangular pulse whose duty cycle is proportional to $u(t)$:

$$D \propto c(t)$$

2.4 Driver Stage

Principle of Operation The PWM signal is applied to the base of a TIP41C NPN power transistor, operating in a common-emitter switching configuration. The transistor switches the motor current on and off according to the PWM duty cycle, acting as the interface between the low-power analog control circuit and the higher-current motor circuit.

- When the PWM signal is high, the base current drives the TIP31C into saturation, connecting the motor across the supply through the collector-emitter path. When the PWM signal is low, the transistor turns off and motor current ceases.
- The average voltage delivered to the motor is approximately:

$$V_{\text{motor, avg}} = D \times V_{\text{supply}}$$

- A flyback diode is connected across the motor terminals to protect the TIP31C from inductive voltage spikes generated by the motor windings when the transistor switches off.
- A base resistor limits the base current to a safe value while ensuring the transistor remains fully saturated at the expected motor current.

2.5 Power Supply

The control circuit is powered from a regulated $\pm 12\text{V}$ dual supply, while the motor is driven from a separate 6V supply rail to prevent motor switching noise from coupling into the sensitive analog PID stages.

3 Component Selection



(a)



(b)



(c)



(d)



(e)



(f)

- (a) **TL071 Op-Amp** – JFET-input operational amplifier with higher slew rate and low noise; used in the PWM generation stage where fast, clean switching transitions are required.
- (b) **LM741 Op-Amp** – General-purpose operational amplifier; used throughout the feedback, error detection, and PID stages where high-speed performance is not critical.
- (c) **2N3906 Transistor** – PNP general-purpose transistor used for switching and signal conditioning in the frequency-to-voltage converter.
- (d) **1N914 Diode** – Fast-switching small-signal diode; used for pulse shaping in the feedback path and waveform clamping in the PWM generator.
- (e) **TIP41C Transistor** – NPN power transistor switching the motor as collector load, base driven by the PWM signal. Supports up to 100V and 6A .
- (f) **N20 DC Motor** – Compact DC motor with a built-in magnetic encoder, used both as the actuator and as the speed-sensing element for the feedback loop.

4 Software Simulation and Hardware Testing

4.1 Simulations

All analog sub-circuits of the PID controller were evaluated using circuit simulation software prior to hardware implementation. **Multisim** is used to verify the transient response, gain accuracy and PWM switching behavior of each block under varying reference and load conditions.

4.1.1 Feedback Generation

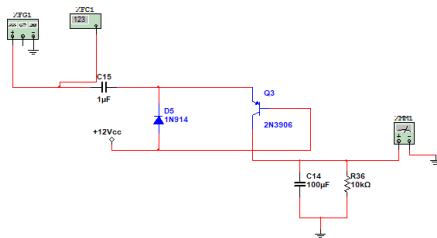


Figure 5: F-V Converter Simulation

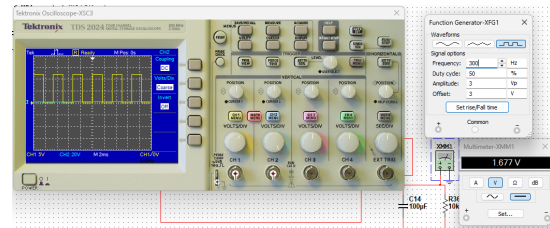


Figure 6: F-V Converter Output Waveform

4.1.2 PID Controller

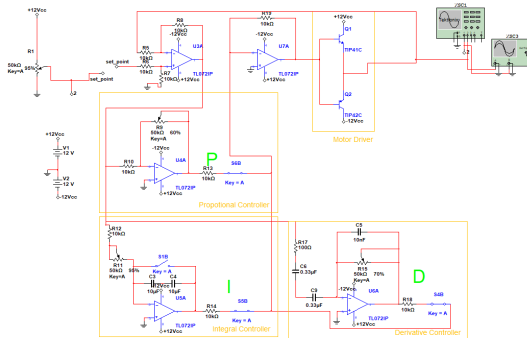


Figure 7: PID Controller Simulation

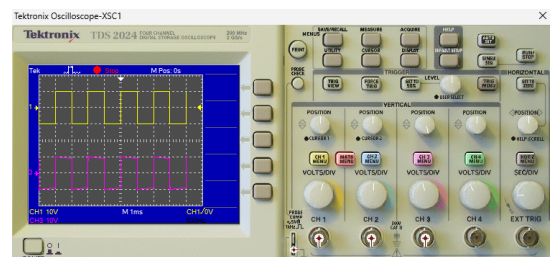


Figure 8: Step Response of the PID Output

4.1.3 PWM Generation

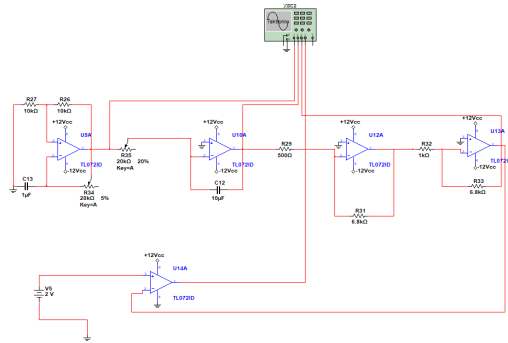


Figure 9: PWM Generator Simulation

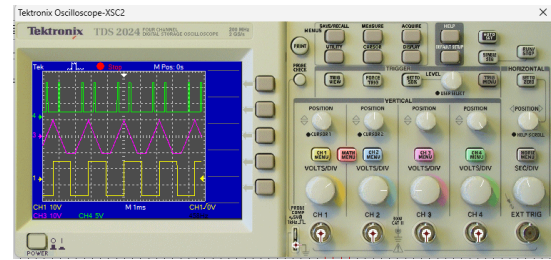


Figure 10: Triangular Carrier and PWM Output Waveforms

4.2 Hardware Testing

4.2.1 Breadboard Implementation

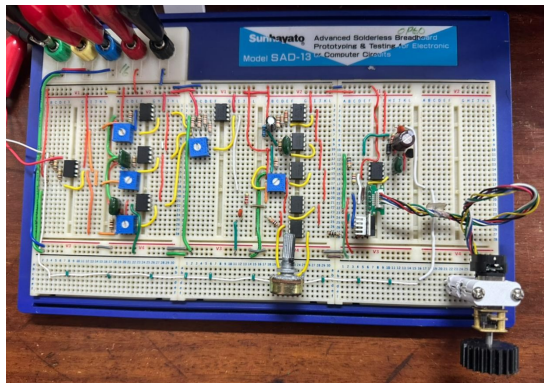


Figure 11: Breadboard Implementation – Close-up

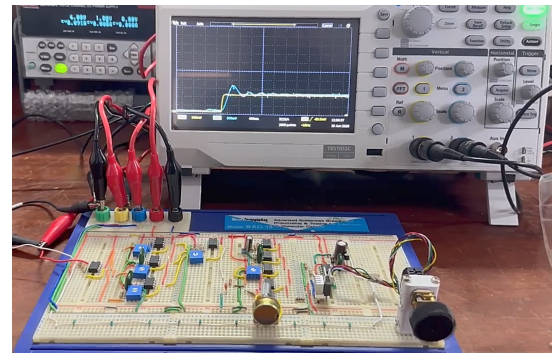


Figure 12: Breadboard Implementation – Overview

4.2.2 Step Response Testing

The system response to a step change in reference speed, and its disturbance rejection under sudden load application, were verified on the oscilloscope.

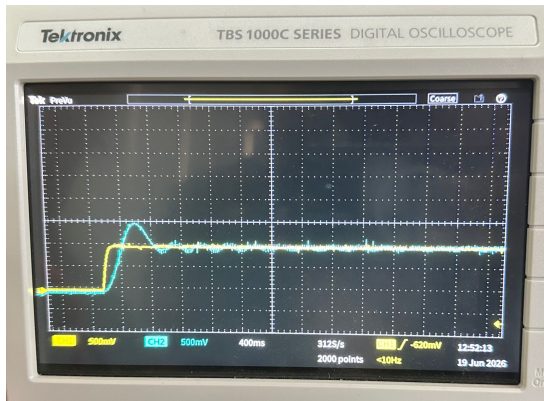


Figure 13: Measured Step Response

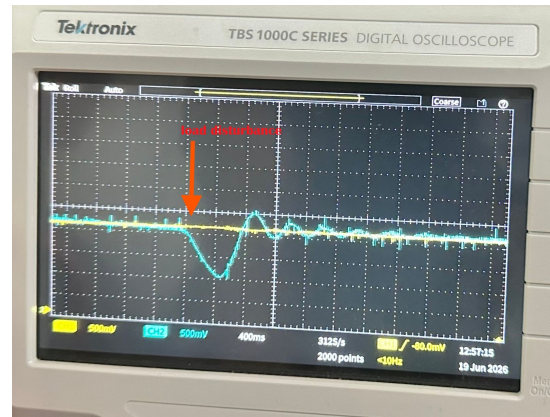


Figure 14: Response to Load Disturbance

Demonstration Videos:

1. Step Response: <https://youtu.be/ZAlQ6-Eypqw>
2. Response to Load Disturbance: <https://youtu.be/ZAlQ6-Eypqw>

5 Performance Parameters

Parameter	Condition	Min	Typical	Max	Units
Supply Voltage (Control)	Operating	N/A	± 12	± 15	V
Motor Supply Voltage	Operating	N/A	6	N/A	V
Reference Speed Range	Linear operation	N/A	N/A	300	rpm
Steady-State Error	Standard load	0	0.1	N/A	%
SNR	Standard load	0.01	0.1	N/A	
Overshoot	Step response	N/A	0.2	N/A	%
Settling Time	Step input	N/A	N/A	1	s
Power Consumption	Nominal	N/A	N/A	6	W
Operating Temperature		N/A	25	N/A	$^{\circ}\text{C}$

Table 1: Performance parameters of the PID motor controller.

6 Conclusion & Future Works

The PID motor controller successfully regulates the speed of a DC motor using a fully analog closed-loop control system. The encoder-based feedback generation, PID compensation network, PWM generation circuit, and motor driver stage were individually verified through simulation and breadboard implementation. The integrated system demonstrated stable operation, effective disturbance rejection, and reduced steady-state error while maintaining the desired motor speed under varying operating conditions.

Future improvements include enhancing the PID controller by incorporating Kalman filtering techniques to improve noise rejection and state estimation accuracy. In addition, the motor driver stage can be upgraded to support bidirectional motor control, enabling the motor to rotate in both clockwise and counterclockwise directions. Further developments may also focus on improving control accuracy and extending the overall operating speed range of the system.

7 Contribution of Group Members

Index Number	Name	Contribution
230147L	Dilhara D.S	Simulation of PID And Breadboard implementation
230724E	Wijesekara W.A.G.S	Simulation of PID And Breadboard implementation
230726L	Wijesinghe U.G.S.K.D	Simulation of PWM Generation And Breadboard implementation
230727P	Wijesinghe W.A.P	Simulation of Feedback generation And Breadboard implementation

Table 2: Contribution of each group member

8 Acknowledgment

We extend our sincere appreciation to Dr.Wageesha Manamperi and Dr.Jayathu Samarawickrama for their invaluable guidance throughout the project. We also acknowledge their feedbacks, which improved our final design. We are grateful for the support from the ENTC academic and non-academic staff throughout the project. Finally, we appreciate the efforts and contributions of all our group members, whose teamwork and dedication made the project successful.

9 References

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